

for MLP (FFN) layers. (only matmul operations)

Start: Flops = 0, bytes-transferred = 0

Step 1: Read X ($B \times T \times D$) From HBM

bytes-transferred $\leftarrow$ input $= 2 * B * T * D$

2 bytes per FP16 element
Read + write accounting

Step 2: Read $W_{up}(D \times F)$, $W_{gate}(D \times F)$, $W_{down}(F \times D)$ from HBM.

bytes-transferred $\leftarrow$ weights $= 3 * 2 * D * F$

Step 3: compute $V = X(B \times T \times D) @ W_{up}(D \times F)$

Flops $\leftarrow 2 * B * T * D * F$ $\rightarrow$ each multiply-add pair operations = 2 Flop

Step 4: Write $V(B \times T \times F)$ to HBM.

hidden activation $\rightarrow$ bytes-transferred $\leftarrow 2 * B * F$

Step 5: compute $G = X(B \times T \times D) @ W_{gate}(D \times F)$

Flops $\leftarrow 2 * B * T * D * F$

Gelu Step 6: Write $G(B \times T \times F)$ to HBM

bytes-transferred $\leftarrow 2 * B * T * F$

Step 7: compute $Y = \text{Gelu}(G) * V(B \times T \times F) @ W_{down}(F \times D)$

outputs $\rightarrow$ Flops $\leftarrow 2 * B * T * D * F$

Step 8: Write $Y(B \times T \times D)$ to HBM.

bytes-transferred $\leftarrow 2 * B * T * D$

To sum up: Flops = $6 * B * T * D * F$

Bytes-transferred = $4 * B * T * D + 4 * B * T * F + 6 * D * F$

intensity = (Flops / bytes-transferred). simplify ()

Assume that $B * T$ is much smaller. D and F

intensity = intensity.subs(D , $c * B * T$).subs(F , $c * B * T$).limit(c , ∞).

$\Rightarrow$ intensity = $B * T$

simplify ()

What is Arithmetic Intensity? (AI)

$\downarrow$
AI =

FLOPS

Bytes-transferred

$\downarrow$ low AI: memory-bound (wait on HBM)
 $\downarrow$ high AI: compute-bound (GPU's ALU are busy)